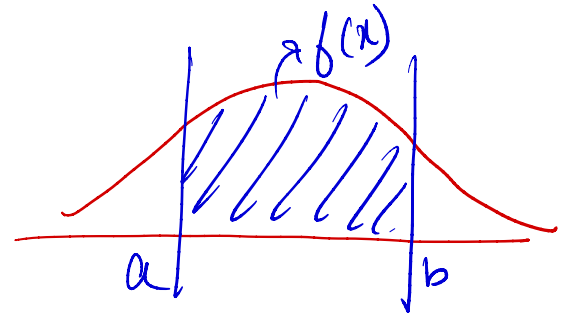


$$\int_a^b f(x) dx$$

Module 3

Chapter 4

Review of Integrals



We learnt that **differential calculus** is the study of the rate of change of quantities.

In differential calculus, the **derivative** $f'(x)$ of a function $f(x)$ is a function that measures the rate of change of $f(x)$.

This chapter covers **integration** of functions.

There are *two types of integration* on a function $f(x)$:

- **Indefinite Integral** of $f(x)$ with respect to x , written as:

$$\int \underline{f(x)} dx = f(x) \\ f'(x) = f(x)$$

- **Definite Integral** of $f(x)$ between two *limits* a and b , written as:

$$\int_a^b f(x) dx$$

What's the difference between Definite and Indefinite Integrals?

- The **indefinite integral** is precisely the *reverse* process of **differentiation**; this result is known as the **Fundamental Theorem of Calculus (FTC)**. It is a **function** and answers the question, "*What function when differentiated gives $f(x)$?*"
- The **definite integral** represents (amongst other things) the **signed area** bounded by the **curve $f(x)$** and the lines $x = a$ and $x = b$. It is a **number** and represents the area under the curve $f(x)$ from $x=a$ to $x=b$.

The **FTC** relates these two integrals in the following manner:

To compute a definite integral, find the antiderivative (indefinite integral) of the function and evaluate at the endpoints $x = a$ and $x = b$.

Indefinite Integrals or Antiderivatives

Given a function $f(x)$, we say $F(x)$ is an antiderivative of $f(x)$ if $F'(x) = f(x)$

NOTE: I used “an antiderivative” and not “the antiderivative”!

Let's understand why.

Which of the following are antiderivatives of $2x$?

- x^2
 - $x^2 + 3$
 - $x^2 - 6$
- $\int 2x dx = x^2 + C$

We can see that there are several choices for antiderivatives of $2x$! Then, how could these antiderivatives differ?

Theorem: (*Antiderivatives differ by a constant*) If $F(x)$ and $G(x)$ are antiderivatives of a function $f(x)$, then there is a constant C such that $F(x) = G(x) + C$.

The collection of all antiderivatives of a function $f(x)$ is denoted by the symbol

$$\int f(x) dx$$

Some common antiderivatives:

Function	antiderivative
$f(x) = a$	$\int f(x) dx = ax + C$
$f(x) = ax^n$	$\int f(x) dx = \frac{ax^{n+1}}{n+1} + C$
$f(x) = ax^{-1}$	$\int f(x) dx = a \ln x + C$
$f(x) = ae^{kx}$	$\int f(x) dx = \frac{1}{k} ae^{kx} + C$

$f(x) = a \cos(kx)$	$\int f(x)dx = \frac{1}{k}a \sin(kx) + C$
$f(x) = a \sin(kx)$	$\int f(x)dx = -\frac{1}{k}a \cos(kx) + C$

Table 1: Some common antiderivatives (throughout a, k, C are constants)

Antiderivative Rules

- **Addition:** $\int (f(x) + g(x))dx = \int f(x)dx + \int g(x)dx$
- **Scalar Multiplication:** $\int (af(x))dx = a \int f(x)dx$

Let's look at an example:

Find $\int (16 - x^2)dx$

- Using the addition rule, we can break it into : $\int 16dx - \int x^2dx$
- We then use the above table to find the corresponding antiderivatives:

$$\int 16dx = 16x + C_1$$

$$\int x^2dx = \frac{x^3}{3} + C_2$$

$$\int 16dx - \int x^2dx = 16x - \frac{x^3}{3} + C_1 - C_2 = 16x - \frac{x^3}{3} + C$$

Let's Practice

Find the following indefinite integrals:

1. $\int \left(\cos(x) - \frac{3}{x^5} \right) dx$
2. $\int (12x^3 - 9x^2 + 2) dx$
3. $\int (8w^7 - 10w^2 + 20) dw$
4. $\int (\sin(2x) + 2\cos(x)) dx$

$$1. \int \left(\cos(x) - \frac{3}{x^5} \right) dx$$

$$\rightarrow \int \cos(x) dx - \int \frac{3}{x^5} dx$$

$$= \sin(x) - \int 3x^{-5} dx$$

$$= \sin(x) - \frac{3}{-4} x^{-4} + C = \sin(x) + \frac{3}{4x^4} + C$$

$$2. \int (12x^3 - 9x^2 + 2) dx$$

$$\rightarrow 12 \int x^3 dx - 9 \int x^2 dx + 2 \int 1 dx$$

$$\rightarrow \frac{12}{4} x^4 - \frac{9}{3} x^3 + 2x + C$$

$$3. \int (8w^7 - 10w^2 + 20) dw$$

$$= \frac{8w^8}{8} - \frac{10w^3}{3} + 20w + C$$

$$= \frac{\cancel{8}w^8}{\cancel{8}} - \frac{10w^3}{3} + 20w + C$$

$$4. \int (\sin(2x) + 2 \cos(x)) dx$$

$$= -\frac{1}{2} \cos(2x) + 2 \sin(x) + C$$

Definite Integrals

Theorem: Let f be continuous on $[a, b]$, If F is an antiderivative of f then

$$\int_a^b f(x)dx = F(b) - F(a)$$

Let's take an example

Find the following definite integral:

$$\int_0^{\ln(3)} 2e^x dx$$

We first calculate the antiderivative and then evaluate the expression using the above theorem:

$$\int_0^{\ln(3)} 2e^x dx = [2e^x]_0^{\ln(3)} = 2e^{\ln(3)} - 2e^0 = 2(3) - 2 = 4$$

Let's Practice

Find the following definite integrals

1. $\int_3^0 (15w^4 - 13w^2 + w) dw$

2. $\int_1^2 \left(\frac{1}{7z} + \frac{\sqrt[3]{z^2}}{4} - \frac{1}{2z^3} \right) dz$ * $\ln(1) = 0$

3. $\int_{-2}^4 \left(x^6 - x^4 + \frac{1}{x^2} \right) dx$

4. $\int_0^4 f(t)dt$ where $f(t) = \begin{cases} 2t & \text{when } t > 1 \\ 1 - 3t^2 & \text{when } t \leq 1 \end{cases}$

$$1. \int_3^0 15w^4 - 13w^2 + w dw$$

$$\rightarrow \left[\left(\frac{15}{5} w^5 - \frac{13}{3} w^3 + \frac{w^2}{2} \right) \right]_3^0$$

$$= -\frac{1233}{2}$$

$$2. \int_1^2 \left(\frac{1}{7z} + \frac{\sqrt[3]{z^2}}{4} - \frac{1}{2z^3} \right) dz \quad * \ln(1) = 0$$

$$= \frac{1}{7} \int_1^2 z^{-1} dz + \frac{1}{4} \int_1^2 z^{2/3} dz - \frac{1}{2} \int_1^2 z^{-3} dz$$

$$= \left[\left(\frac{1}{7} \ln|z| + \frac{1}{4(5/3)} z^{5/3} - \frac{1}{2(-2)} z^{-2} \right) \right]_1^2$$

$$= \left[\left(\frac{1}{7} \ln|z| + \frac{3}{20} z^{5/3} + \frac{1}{4z^2} \right) \right]_1^2$$

$$= \left(\frac{1}{7} \ln(2) + \frac{3}{20} \sqrt[3]{2^5} + \frac{1}{16} \right) - \left(\frac{1}{7} \ln(1) + \frac{3}{20} + \frac{1}{4} \right)$$

$$= \frac{1}{7} \ln(2) + \frac{3}{20} \sqrt[3]{2^5} - \frac{27}{80}$$

* 3. $\int_{-2}^4 x^6 - x^4 + \frac{1}{x^2} dx$

→ Here our range of integration is $[-2, 4]$

Note that our 3rd term $\rightarrow \frac{1}{x^2}$ is undefined

at $x=0$. And 0 is a part of the range

we are integrating over.

∴ The function is not continuous on this interval.

∴ Integral cannot be done.

4. $\int_0^4 f(t) dt$ where $f(t) = \begin{cases} 2t & t > 1 \\ 1-3t^2 & t \leq 1 \end{cases}$



→ graph for $\frac{1}{x^2}$

→ since the function changes at $t=1$,
we need to break our integral down.

$$\therefore \int_0^4 f(t) dt = \int_0^1 f(t) dt + \int_1^4 f(t) dt$$

$$= \int_0^1 (1 - 3t^2) dt + \int_1^4 2t dt$$

$$= \left[t - t^3 \right]_0^1 + \left[t^2 \right]_1^4$$

$$= 15$$

Techniques for special cases

Preliminary Algebra

A basic technique you can apply when solving for antiderivatives, is using preliminary algebra to reduce the integrand into a simpler form.

Let's take an example

Let's find the antiderivative:

$$\int (7 - x^2)^2 dx$$

We can simplify the integrand:

$$\int (7 - x^2)^2 dx = \int (49 - 14x^2 + x^4) dx$$

You can now easily find the antiderivative:

$$\int (49 - 14x^2 + x^4) dx = 49x - 14 \frac{x^3}{3} + \frac{x^5}{5} + C$$

Method of Substitution

Sometimes the integrand may be complex and just using preliminary algebra might not help. In such cases, a change of variable can often help. Often, when an integrand involves a composition of a trigonometric, exponential, or power and another function, letting a new variable represent the inner function helps!

Let's take an example:

Let's find the antiderivative:

$$\int 2xe^{(x^2+1)} dx$$

We can see that no algebra will help us simplify the above integrand!

We have one antiderivative formula that involves exponents: $\int e^x dx = e^x + C$

(It looks like we have a function of a function in our integrand!)

Let's start by letting $x^2 + 2$ be a new variable, say, u .

We will now try to write everything in the integrand in terms of the new variable.

We know,

$$\frac{du}{dx} = \frac{d(x^2+1)}{dx} = 2x$$

$$\begin{aligned} x^2 + 1 &= u \\ \frac{du}{dx} &= 2x \\ du &= 2x dx \end{aligned}$$

$$\int 2x e^{(x^2+1)} dx \rightarrow \int e^u du$$

So,

$$du = 2x dx$$

We can now rewrite our antiderivative as:

$$\int e^u du = e^u + C = e^{x^2+1} + C$$

$$\begin{aligned} &= e^u + C \\ &= e^{(x^2+1)} + C \end{aligned}$$

Integration by parts

$$\int f g dx$$

Sometimes we have an integrand which is the product of two function and neither Preliminary Algebra nor integration by substitution works! Sometimes we need use "Integration by parts". Let's first derive the formula and look at an example to understand this better!

We know, according to product rule for differentiation,

$$(fg)' = f'g + g'f$$

$$\begin{aligned} &u \quad v \\ &\int f g dx \\ &\int u dv \end{aligned}$$

Taking the integral of both sides:

$$\int (fg)' dx = \int (f'g + g'f) dx$$

$$fg = \int f'g dx + \int g'f dx$$

$$\int g' f dx = f g - \int f' g dx$$

This doesn't seem any better! Let's do some substitutions

Let

$$u = f(x), v = g(x)$$

So,

$$\frac{du}{dx} = f'(x)$$

$$du = f'(x) dx$$

Similarly,

$$\frac{dv}{dx} = g'(x)$$

$$dv = g'(x) dx$$

$$\begin{array}{c} \int f g dx \\ \downarrow \\ \int u dv \end{array}$$

Using these substitutions, we can rewrite the formula $\int g' f dx = f g - \int f' g dx$ as:

$$\int u dv = uv - \int v du$$

To use this formula, we will need to identify ***u*** and ***dv***, compute ***du*** and ***v*** and then use the formula.

It won't always be clear what the correct choices are and we will, on occasion, make the wrong choice. This is not something to worry about. If we make the wrong choice, we can always go back and try a different set of choices!

This does lead to the obvious question of *how do we know if we made the correct choice for u and dv* ? The answer is actually pretty simple: *We made the correct choices for u and dv if, after using the integration by parts formula the new integral (the one on the right of the formula) is one we can actually integrate.*

$$\int u dv$$

Let's take an example:

Let's evaluate the following:

$$\int x e^{6x} dx$$

→ idea is to write this down as $\int u dv$

If we just had an x by itself or e^{6x} by itself we could do the integral easily enough.

Likewise, if the integrand was $x e^{6x^2}$ we could do the integral with a substitution.

Unfortunately, however, neither of these are options.

So, we resort to integration by parts!

Note that in doing integration by parts anything that we choose for u *will be differentiated*. So, it seems that choosing $u = x$ will be a good choice since upon differentiating the x *will drop out*.

Now that we have our u , dv will be everything that remains!

So,

$$u = x, dv = e^{6x} dx$$

$$\underline{du = dx}, v = \int dv = \int e^{6x} dx = \boxed{\frac{1}{6} e^{6x}}$$

Our integral now becomes:

$$\int x e^{6x} dx = \int u dv$$

We can now easily use the integration by parts formula:

$$\int u dv = uv - \int v du$$

$$\int x e^{6x} dx = \frac{x}{6} e^{6x} - \int \frac{1}{6} e^{6x} dx = \frac{x}{6} e^{6x} - \frac{1}{36} e^{6x} + C$$

Let's Practice

Find the following integrals:

1. $\int_{-2}^0 (2t^2 \sqrt{1-4t^3}) dt$
2. $\int_{-1}^5 (1+w)(2w+w^2)^5 dw$
3. $\int_0^1 3(4x+x^4)(10x^2+x^5-2)^6 dx$
4. $\int \sqrt{x(x+1)^2} dx$
5. $\int \frac{2x^2-3}{2x^2} dx$
6. $\int (3t+5) \cos\left(\frac{t}{5}\right) dt$

Integration by substitution

$$\int_{-2}^0 2t^2 \sqrt{1-4t^3} dt$$

$\sqrt{1-4t^3}$ is the composite part of the function
(Taking the internal function as u)

$$\text{let } u = 1 - 4t^3$$

$$\frac{du}{dt} = -12t^2$$

$$\Rightarrow du = -12t^2 dt$$

$$\Rightarrow \frac{-1}{12} du = t^2 dt$$

(Trying to write the rest of the terms in terms of u, du)

$$\int -\frac{2}{12} du \cdot \sqrt{u} = \frac{-1}{6} \int \sqrt{u} du = \frac{-1}{6} \int u^{1/2} du$$

$$= -\frac{1}{6} \times \frac{2}{3} \cdot u^{3/2} + C = \left(-\frac{1}{9} u^{3/2} \right) + C$$

* Remember, the limits are in terms of t .
So, convert the limits in terms of u .

$$\left[-\frac{1}{9} (1-4t^3)^{3/2} \right]_{-2}^0$$

→ Converting the limits in terms of t

$$\text{when } t = 0, \quad u = 1$$

$$\text{when } t = -2, \quad u = 33$$

$$\underline{u = 1 - 4t^3}$$

∴ Our actual problem now becomes:

$$\int_{-2}^0 2t^2 \sqrt{1 - 4t^3} dt = -\frac{1}{6} \int_{33}^1 u^{1/2} du = \left[\left(\frac{-1}{9} u^{3/2} \right) \right]_{33}^1$$
$$= \frac{1}{9} (33 \sqrt{33} - 1)$$

$$\int_0^1 3(4x + x^4)(10x^2 + x^5 - 2)^6 dx$$

Integration by substitution

$$u = 10x^2 + x^5 - 2$$

$$\frac{du}{dx} = 20x + 5x^4 = 5(4x + x^4)$$

$$\frac{du}{5} = (4x + x^4) dx$$

changing our integral limits in terms of u

$$\text{when } x = 0, \quad u = -2$$

$$\text{when } x = 1, \quad u = 9$$

$$\therefore \int_0^1 3(4x + x^4)(10x^2 + x^5 - 2)^6 dx$$

$$= \int_{-2}^9 \frac{3}{5} u^6 du = \frac{3}{5} \int_{-2}^9 u^6 du$$

$$= \left[\left(\frac{3}{35} u^7 \right) \right]_{-2}^9 = \frac{14349291}{35}$$

$$\int_{-1}^5 (1+w)(2w+w^2)^5 dw \quad \text{Integration by substitution}$$

$$\rightarrow 2w + w^2 = u$$

we need to change the limits of the integration in terms of u

$$\text{when } w = 5, \quad u = 2(5) + (5)^2 = 35$$

$$\text{when } w = -1, \quad u = 2(-1) + (-1)^2 = -1$$

$$\frac{du}{dw} = 2 + 2w$$

$$du = 2(1+w)dw \Rightarrow \frac{du}{2} = (1+w)dw$$

$$\Rightarrow \int_{-1}^5 (1+w)(2w+w^2)^5 dw = \int_{-1}^{35} \frac{1}{2} (u)^5 du$$

$$= \left[\left(\frac{1}{2} \times \frac{1}{6} u^6 \right) \right]_{-1}^{35}$$

$$= \left[\frac{1}{12} u^6 \right]_{-1}^{35} = 153188802$$

Integration by parts

$$\int \sqrt{x} (x+1)^2 dx$$

OR

simple algebra:

$$\int x^{1/2} (x+1) dx$$

$$(x+1) = u$$

$$\frac{du}{dx} = 1$$

$$du = dx$$

$$dv = x^{1/2} dx$$

$$v = \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} = \frac{2x^{3/2}}{3}$$

$$\int x^{1/2} (x+1) dx$$

$$= \int x^{3/2} dx + \int x^{1/2} dx$$

$$= \frac{x^{\frac{3}{2}+1}}{\frac{3}{2}+1} + \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} + C$$

$$= \frac{2}{5} x^{5/2} + \frac{2}{3} x^{3/2} + C$$

$$\int u dv = uv - \int v du$$

$$\int x^{1/2} (x+1) dx = \frac{2(x+1)x^{3/2}}{3} - \int \frac{2x^{3/2}}{3} dx$$

$$= \frac{2}{3} (x+1) x^{3/2} - \frac{2}{3} \cdot \frac{x^{\frac{3}{2}+1}}{\left(\frac{3}{2}+1\right)} + C$$

$$= \frac{2}{3} x^{3/2} (x+1) - \frac{4}{15} x^{5/2} + C$$

$$= \frac{2}{3} x^{5/2} + \frac{2}{3} x^{3/2} - \frac{4}{15} x^{5/2} + C$$

$$= \frac{2}{5} x^{5/2} + \frac{2}{3} x^{3/2} + C$$

$$\int \frac{2x^2 - 3}{2x^2} dx \quad \text{simple algebra}$$

$$\int \frac{2x^2}{2x^2} dx - \int \frac{3}{2x^2} dx$$

$$= x + C - \frac{3}{2} \int x^{-2} dx$$

$$= x - \frac{3 \cdot x^{-1}}{-2} + C$$

$$= x + \frac{3}{2x} + C$$

$$\int (3t+5) \cos\left(\frac{t}{5}\right) dt \quad \text{Integration by parts}$$

$$3t+5 = u$$

$$\frac{du}{dt} = 3$$

$$du = 3dt$$

$$dv = \cos\left(\frac{t}{5}\right) dt \Rightarrow \int dv = \int \cos\left(\frac{t}{5}\right) dt$$

$$v = 5 \sin\left(\frac{t}{5}\right)$$

$$\therefore \int u dv = uv - \int v du$$

$$= 5(3t+5) \sin\left(\frac{t}{5}\right) - \int 5 \sin\left(\frac{t}{5}\right) \cdot 3 dt$$

$$= 5(3t+5) \sin\left(\frac{t}{5}\right) + 75 \cos\left(\frac{t}{5}\right) + C$$

Some Extra practice!

1. $\int x^2 e^x dx$

2. $\int x \ln x dx$

$$1. \int x^2 e^x dx$$

$$x^2 = u$$

$$\frac{du}{dx} = 2x$$

$$du = 2x dx$$

$$dv = e^x dx$$

$$v = \int dv = \int e^x dx = e^x$$

$$\int u dv = uv + \int v du$$

$$\therefore \int x^2 e^x dx = x^2 e^x - \underbrace{\int e^x 2x dx}_{\downarrow} + C$$

$$= 2 \int x e^x dx \rightarrow \text{using integration by parts}$$

$\downarrow$

$$x = u$$

$$\therefore \frac{du}{dx} = 1 \rightarrow du = dx$$

$$e^x dx = dv$$

$$v = \int dv = \int e^x dx = e^x$$

$$\therefore 2 \int x e^x dx = 2 \left[x e^x - \int e^x dx \right]$$

$$= 2x e^x - 2e^x$$

$$= 2e^x (x - 1)$$

$$\Rightarrow \int x^2 e^x dx = x^2 e^x - 2e^x (x - 1) + C$$

$$= e^x (x^2 - 2x + 2) + C$$

$$2. \int x \ln x \, dx$$

$$u = \ln x \quad \Rightarrow \quad \frac{du}{dx} = \frac{1}{x} \quad \Rightarrow \quad du = \frac{dx}{x}$$

$$dv = x \, dx \quad \Rightarrow \quad \int dv = \int x \, dx = \frac{x^2}{2}$$

$$\therefore \int x \ln x \, dx = \frac{x^2}{2} \ln x - \int \frac{x^2}{2} \cdot \frac{dx}{x}$$

$$= \frac{x^2}{2} \ln x - \int \frac{x}{2} \, dx$$

$$= \frac{x^2}{2} \ln x - \frac{x^2}{4} + C$$

OR

$$u = x \quad du = dx$$

$$dv = \ln x \, dx \quad \Rightarrow \quad \int dv = \int \ln x \, dx$$

↳ solve using integration by parts

$$v = x \ln x - x$$

$$\ln x = u$$

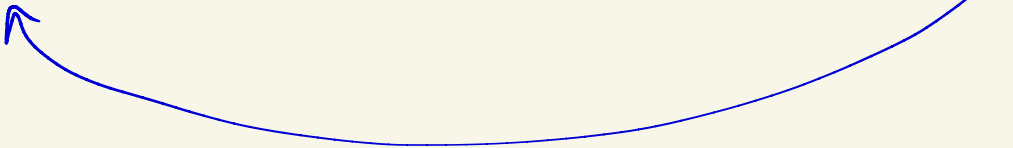
$$du = \frac{1}{x} dx$$

$$dx = du \Rightarrow v = x$$

$$\Rightarrow \int \ln x \, dx = x \ln x - \int \frac{x}{x} \, dx \\ = x \ln x - x + C$$

$$\int x \ln x dx = x(x \ln x - 1) - \int x(\ln x - 1) dx$$

$$\int x \ln x dx = x^2 \ln x - x - \int x \ln x dx + \int x dx$$


$$2 \int x \ln x dx = x^2 \ln x - x + \frac{x^2}{2} + C$$

$$\int x \ln x dx = \frac{x^2}{2} \ln x - \frac{x}{2} + \frac{x^2}{4} + C$$

$$= \frac{x^2}{2} \ln x - \frac{x^2}{4} + C$$